1 - Course Introduction

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Agenda

1. Course overview
2. Data driven decision making in business

1. Course Overview

Instructors

Lectures: Filipa Reis (filipareis@ucp.pt)

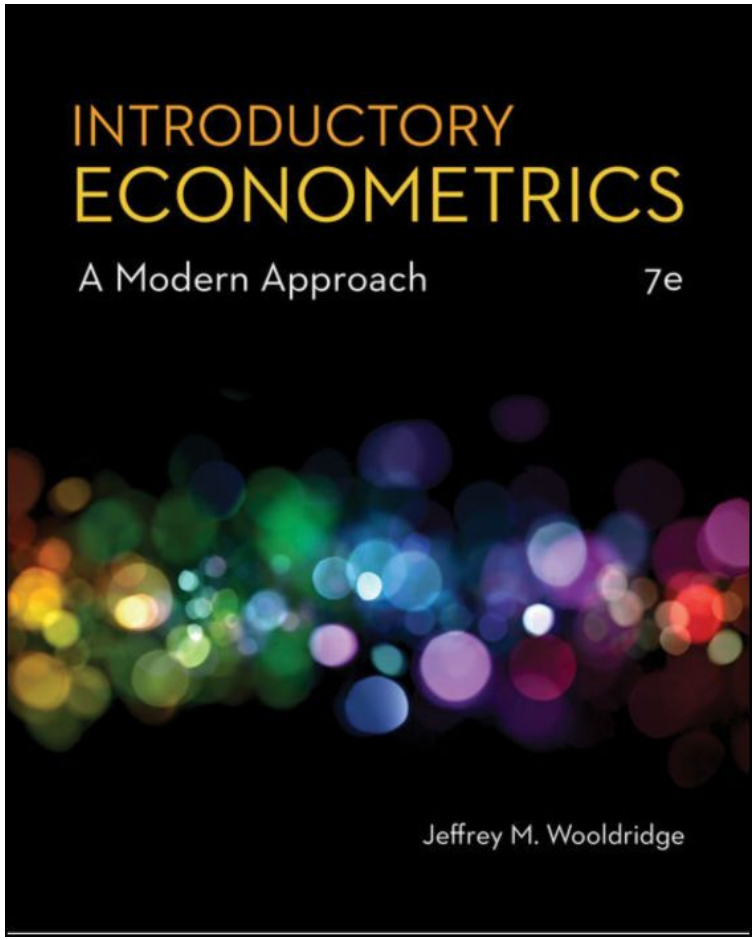
- Mondays (11h00 to 12h20) and Wednesdays (11h00 to 12h20), Room 527
- Office hours: availability listed on Moodle
- Office: 5306 "Rotunda", 3rd floor

Labs: Ana Guedes (amguedes@ucp.pt)

- Fridays from 8h00 to 9h20, Room 527
- Office hours: availability listed on Moodle

Textbook

- JWB: Jeffrey Wooldridge, Introductory Econometrics: A Modern Approach, 7 ed. Thomson.



Course Objectives

This course focuses on the methods and tools most relevant for conducting **applied** research using **non-experimental data**.

1. Understanding the fundamental econometric and statistical concepts and tools.
2. Formulating and testing economic/business hypotheses.
3. Conducting simple and multiple regression for business/economic analysis and policy evaluation.
4. Identifying and mitigating observational data challenges and addressing common regression issues.
5. Interpreting econometric results for informed decision-making.
6. `R` programming.

Course Contents

1. The Nature of Econometrics and Economic Data
2. Fundamentals of Probability
3. Fundamentals of Mathematical Statistics
4. Regression Analysis with Cross-Sectional Data
5. Advanced Topics in Linear Regression
6. Simple Panel Data Methods
7. Advanced Panel Data Methods

Grading

The final grade is based on:

- Individual Assignments (10%)
- Group Project (40%)
- Midterm and Final Exam (50%, 25% each) – minimum avg. grade is 6.5.

2. Data-Driven Decision Making in Business

What is Econometrics for Business?

Econometrics is the application of **statistical methods** to **economic data** to develop theories or test hypotheses.

For **business analytics**, it involves:

- Estimating business-related relationships.
- Evaluating and implementing business strategies and policies.
- Forecasting business and economic variables.

Econometrics helps in making **data-driven decisions by analyzing nonexperimental (observational) data**, often derived from business operations, sales, marketing campaigns, and financial transactions.

Steps in Empirical Business Analysis (1/3)

Empirical analysis in business uses **data** to test hypotheses or estimate relationships that are crucial for decision-making.

Steps:

1. Clearly define a **business/research question**.
    - "How much should we increase our digital marketing budget to maximize sales?"
    - "What is the impact of personalized email campaigns on customer retention rates?"
    - "How does employee productivity change after attending a new training program?"

Steps in Empirical Business Analysis (2/3)

Empirical analysis in business uses **data** to test hypotheses or estimate relationships that are crucial for decision-making.

Steps:

1. Clearly define a **business/research question**.
   - e.g., How much should we increase our digital marketing budget to maximize sales?
2. Develop a formal **business/economic model**.
   - e.g., a simplified description of how different variables like marketing spend, pricing, and customer engagement impact sales.
3. Specify an **econometric model**:

$$Sales = \beta_0 + \beta_1.MarketingSpend + \beta_2.Price + \beta_3.CustomerEngagement + u$$

Steps in Empirical Business Analysis (3/3)

$$Sales = \beta_0 + \beta_1.MarketingSpend + \beta_2.Price + \beta_3.CustomerEngagement + u$$

4. Formulate **research hypotheses**.
    - e.g., Increasing marketing spend by 10% will increase sales by 5%
5. Collect relevant **data**.
    - e.g., sales data, marketing spend data, customer engagement metrics.
6. Use econometric methods to **estimate** the **parameters** and **test** the **hypotheses**.

Causality in Business Decisions

- In business, understanding the **causal effect** of one variable on another is crucial for strategic decision-making.
- The principle of **ceteris paribus** ("all other things being equal") helps isolate the impact of one factor by holding others constant.
- It isn't always possible to literally hold all else equal.
- We will need to consider: have enough other factors been held fixed to make a case for causality?
- It is useful to describe how an **experiment** would have to be designed to infer the causal effect in question.

Impact of Customer Loyalty Program (1/2)

“How does our customer loyalty program affect repeat purchase behavior?”

Impact of Customer Loyalty Program (2/2)

"How does our customer loyalty program affect repeat purchase behavior?"

- **Implicit assumption:** All other factors affecting repeat purchases are held constant.
  - e.g., consumer demographics,
  - consumer preferences,
  - marketing campaigns, etc.
- Could we run an **experiment**? What would be a possible **experimental design**?
- What would be the key **challenges**?

Impact of Remote Work on Productivity (1/2)

"How does remote work impact employee productivity?"

Impact of Remote Work on Productivity (1/2)

"How does remote work impact employee productivity?"

- **Implicit assumption:** All other factors influencing productivity are held constant.
  - e.g., job specific factors,
  - team dynamics,
  - social and economic context, etc.
- Could we run an **experiment**? What would be a possible **experimental design**?
- What would be the key **challenges**?

Measuring the return to education (1/2)

"What is the impact of an additional year of education on a person's wage?"

Measuring the return to education (2/2)

"What is the impact of an additional year of education on a person's wage?"

- **Implicit assumption:** All other factors that influence wages are held fixed.
  - e.g., experience,
  - family background,
  - ability, etc.
- Could we run an **experiment**? What would be a possible **experimental design**?
- What would be the key **challenges**?

Why Econometrics Matters in Business Analytics

- Provides a **framework** for addressing business questions while accounting for the complexity of real-world data.
- Helps **identify** and **quantify** the impact of key business drivers.
- **Controls** for confounding factors to isolate the true effect of individual variables.
- **Validates** business strategies and theories through scientific methods.
- **Predicts** future business outcomes based on historical data.
- **Guides** decision-making by providing evidence-based insights.

Recommended readings

- Wooldridge, *Introductory Econometrics*, Chapter 1